# ST. JOSEPH'S COLLEGE OF ENGINEERING

*(An Autonomous Institution) • OMR, Chennai - 119*

LAB RECORD

## Professional English Lab Integrated

**CLASS CODE: EN25C09**

### STUDENT DETAILS

**Student Name:** _____

**Roll No:** _____

**Date:** _____

**Subject Name:** _____

DEPARTMENT OF HUMANITIES & SCIENCES • AFFILIATED TO ANNA UNIVERSITY • CHENNAI

# Self Introduction: Written Script and One-Page Profile

## Inba Ilakiyan

Aspiring Principal Cloud Architect and eventually Director of Engineering

| | |
|---|---|
| **Links** | GitHub: https://github.com/DubberRuckky/<br>LinkedIn: https://www.linkedin.com/in/inba-ilakiyan-a-e-51909742a |
| **Summary** | I’m hungry for knowledge. When a problem gets in my way, I adapt and improvise until it’s solved.<br><br>My real ambition is game development. Infrastructure work or cloud architecture engineering looks like the best way to build the skills that get me there. Those jobs are also unlikely to be taken over by AI, since they need more situational awareness than an algorithm can easily replicate. |
| **Introduction** | Hi, I’m Inba Ilakiyan, a Computer Science Engineering student (batch 2026–2030) and a Linux enthusiast who likes building things in Python. I’m a strong problem solver, and I like to think a challenge all the way through. Down the line, I want to grow into a Principal Cloud Architect and, eventually, a Director of Engineering. |
| **Skills** | Python, C, and Linux |
| **Experience** | 1st Place in the Mazebot Challenge, Sathyabama Technosummit ‘26 |
| **Education** | CSE, 1st Year |
| **Awards** | Overall Shield Winner (Regional Level) in AWIM 2018 |

# Impressive Personality: Biographical Note

Gabe Newell (“Gaben”), co-founder and president of Valve

Gabe Logan Newell, better known by his nickname Gaben, was born on November 3, 1962. He is an American video game developer and businessman, and he co-founded Valve, the video game company where he serves as president.

Newell was born in Aspen, Colorado, and grew up in Davis, California. In the early 1980s he enrolled at Harvard University but dropped out to join Microsoft. There he helped build the first versions of the Windows operating system, and he also led the port of Doom to Windows 95.

In 1996 he and Mike Harrington left Microsoft and founded Valve, paying for the development of its first game, Half-Life (1998). Harrington sold his stake to Newell and left in 2000. Newell then drove the development of Steam, Valve’s digital distribution service. It launched in 2003, and by 2011 it controlled most of the market for downloaded PC games.

Forbes estimated that Newell owned at least half of Valve as of 2025. He also owns the marine research organization Inkfish and the neuroscience company Starfish Neuroscience. Oceanco, a custom yacht manufacturer, is his too. His estimated net worth was $11 billion as of 2025, which makes him one of the wealthiest people in the United States and the richest person in the video games industry.

## Microsoft

While he was at Harvard, Newell visited his brother at Microsoft, which was not yet a major software developer. Steve Ballmer, then a Microsoft executive, suggested he drop out, and he did, starting work at the company. Newell said later that Microsoft was the best place in the world to learn programming at the time.

He spent 13 years there as a programmer and technical executive and produced the first three releases of Windows. In late 1995, Doom, the 1993 first-person shooter from id Software, was estimated to be installed on more computers worldwide than Microsoft’s new Windows 95. Newell recalled: “[id] ... didn’t even distribute through retail, it distributed through bulletin boards and other pre-internet mechanisms. To me, that was a lightning bolt. Microsoft was hiring 500-people sales teams and this entire company was 12 people, yet it had created the most widely distributed software in the world. There was a sea change coming.” At Microsoft he led the Doom port for Windows 95, which is credited with helping make Windows a viable game platform.

## Valve

Michael Abrash had left Microsoft to work on Quake at id, and that inspired Newell. He and another employee, Mike Harrington, left Microsoft and founded Valve on August 24, 1996. Newell chose founding a company over retiring. He felt it would be more fun to work with “other really smart, motivated, socially orientated people to create product that would affect millions of other people.”

Newell and Harrington paid for the first Valve game themselves, the first-person shooter Half-Life (1998), which was a critical and commercial success. In 2000 Harrington sold his stake in Valve to Newell. For Half-Life 2 (2004), Newell set no deadline and gave Valve a “virtually unlimited” budget, promising to fund it himself if he had to. Valve came close to bankruptcy during a legal battle with Vivendi Games. It ended when an intern found an email showing that Vivendi was destroying evidence.

While Half-Life 2 was in development, Newell spent months building Steam, a digital distribution service for games. By 2011 Steam held between 50% and 70% of the market for downloaded PC games and brought in most of Valve's revenue. At a technology conference in Seattle that year, Newell argued that the best answer to software piracy is a better product, not anti-piracy technology, and he pointed to Steam's success in Russia, where piracy is rife. After the difficult development of Half-Life 2, he said he became "obsessed" with improving Valve's work-life balance.

In 2007 Newell voiced his displeasure about developing for game consoles and called the process for Sony's PlayStation 3 a "waste of everybody's time". On stage at Sony's keynote at E3 2010, he acknowledged that criticism, talked about the open nature of the PlayStation 3 and announced a port of Portal 2, remarking that with Steamworks support it would be the best version on any console. He also called the Xbox Live service a "train wreck" and described Windows 8 as a threat to the open nature of PC gaming. At LinuxCon in 2013, he said Linux and open source development were "the future of gaming", and he accused the proprietary systems of Microsoft and Apple of stifling innovation through slow certification processes.

IGN named Newell one of its top 100 game creators in 2009, writing that it was "almost impossible to gauge" Valve's influence on game design, technology and the video games industry. Forbes listed him in December 2010 as "A Name You Need to Know", mainly for his work on Steam and his partnerships with several major developers. In 2013 he entered the Academy of Interactive Arts & Sciences Hall of Fame and received the BAFTA Fellowship for his contributions to the industry.

Josh Weier, the project lead for Portal 2 (2011), said Newell became less involved in game development after that game, because many developers would blindly agree with his ideas given his position. By 2024, Newell had stepped back further from Valve and was spending more time on personal projects.

## Other ventures

Around 2013, Newell co-founded and funded Foundry10, an educational research organization. In 2018 he donated $20 million to the American artificial intelligence company OpenAI and acted as an informal advisor. In 2022 he and Philip Sabes co-founded the neuroscience company Starfish Neuroscience to develop neural interfaces. In May 2025, Starfish announced that its first chip would be released late that year.

Newell owns Inkfish, the marine research organization, which has several ships and submarines. In November 2022 Inkfish bought the Hadal Exploration System, a private deep-sea exploration platform, from the undersea explorer Victor Vescovo. In August 2025 Newell acquired the privately owned yacht manufacturer Oceanco.

## Charity work

In 2020, Newell and Valve employee Yahn Bernier created a car racing team, the Heart of Racing, to raise money for children's charities in Seattle and New Zealand. That same year Newell worked with Wētā Workshop and Rocket Lab to send a gnome figure from Half-Life 2: Episode Two into space. He donated $1 for every person who watched the launch video within 24 hours. The money went to the pediatric intensive care unit at the Starship Children's Hospital in Auckland, New Zealand.

# Summary of Lecture Notes

NPTEL: Enhancing Soft Skills and Personality (Prof. T. Ravichandran, IIT Kanpur)

## Week 1: Course Introduction and Growth Mindset

The course opens with two preview lectures that recap an earlier course, “Developing Soft Skills”. That course dealt with perception, self-management, excellence, conflict, stress, habits, listening, telephone skills, technology, communication barriers, non-verbal communication, presentations, reading and trust. The big new topic here is mindset: a person’s habitual attitudes and ways of reading a situation.

In the shoe-salesman story, nobody wears shoes. Is that “no chance” or a “splendid opportunity”? The same situation draws opposite responses. Drawing on Carol Dweck, the lectures set a fixed mindset against a growth mindset. People with a fixed mindset avoid challenges, resist change, dwell on their limits, reject criticism and stop learning once school is over. Growth-mindset people welcome challenges, learn from failure and treat learning as lifelong.

Learning mindsets form early. A belief like “I’m not smart enough” limits a person, but it isn’t a fact about their ability. The suggested “secrets” are to behave like a baby (open and unafraid), put up with critical people, reject negative stereotypes and stereotype threat, and look for a sense of belonging. The Hawthorne effect is offered as evidence that belonging and attention motivate people more than incentives do.

## Week 2: Time Management, Procrastination and Assertiveness

Time can’t really be managed. What we manage is our habits and our perception of it. Monochronic people stick to rigid schedules, while polychronic people stay flexible. Others tend to judge our personality by our punctuality and by how we behave under deadline pressure. To control time, log it, look for “time leakage” (interruptions, indecision, poor organisation, social media, unmanaged email) and make room for productive work. Working at your biological peak time helps. Covey’s urgent/important quadrants and the 80/20 Pareto principle help with prioritising, and “time stealers” such as waiting and travel can be turned into “time gifters.”

The lectures trace procrastination to Parkinson’s law, fear, laziness, perfectionism and addiction, along with the plain feeling of being overwhelmed. The remedies are:

- Eat the elephant bite by bite.
- Do the “frog” (the hardest task) first.
- Use the Kaizen one-minute principle.
- Keep a clean workspace.
- Set goals you actually want and declare them publicly.
- Reward yourself.
- Use the Zeigarnik effect: once you start, you feel compelled to finish.

The week ends with a first look at assertiveness, or not saying “yes” when you mean “no.”

## Week 3: Assertiveness, Anger and Positive Thinking

The lectures compare passive, assertive and aggressive people on body language, listening, feelings and outcomes. Assertiveness means stating your rights without violating anyone else's, and it ideally ends in a win-win. Practical steps include checking your relationships, dropping self-debilitating thoughts, avoiding discouraging people, following Stanley and Nancy's commandments and learning four ways to say no.

On anger, Aristotle's point is that the hard part is being angry at the right person, to the right degree, at the right time. Techniques include postponing your reaction, counting, deep breathing, physical exertion, Lincoln's unsent letter, silence and channelling anger into creative work. The lectures then turn to positive thinking through the story of two hospital patients and a "window" that faced a blank wall. Changing negative thinking takes desire and determination, introspection and discarding limitations. It also means building a positive self-image through repetition and visualisation, and bringing body, mind and world into sync.

## Week 4: People Skills and Being Attractive

Forty traits make others dislike you, and their opposites make people like you. Examples include egotism, aggression, pessimism, fault-finding, gossiping, false promises, moodiness and favouritism, set against selflessness, optimism, giving credit, politeness, trustworthiness, honesty and humility. For dealing with gossip, Socrates' three-filter test (truth, goodness, utility) is suggested.

Attractiveness has physical, mental, emotional and spiritual layers. Physically, it means neat footwear, a fit body, good diet and sleep, and a signature personality. Mentally, it means intellectual charm and a touch of mystery. Emotionally, it means remembering names, valuing yourself and being yourself. Spiritually, it comes down to mindfulness, deeper connection, helping without expectation and "paying it forward."

## Week 5: English Skills and Common Errors

English is the global lingua franca, and avoiding common errors is presented as part of soft skills. Using Sonia Sharma's Common Errors in English, the lectures work through repeated self-scored exercises and explanations. Topics include confused word pairs (resume/résumé, principal/principle, alone/lonely), Indianisms (passed out, joint family, cousin brother) and subject-verb agreement (collective, uncountable and plural-form nouns). The lectures also cover prepositions (since/for, at/in, by/on), articles, collocations and word choices like weather/climate. Twain and Branson are quoted on learning by exposure rather than by memorising rules.

## Week 6: Humour and Money

Humour reduces stress, improves health and helps communication. It works at three levels: a smile, hearty laughter, and sarcasm, which is best avoided. Adults laugh far less than children (about 4 times a day versus 300). At work, humour resolves conflict, boosts creativity and productivity, and lets you speak the truth lightly. Suggestions include a joke board, a humour club and a humour hall.

The focus then turns to money and personality. How you treat rich and poor people, how you borrow and lend, and how you handle trust all reveal emotional maturity and integrity. Wealth-building advice includes tracking income and expenses, separating need from greed, avoiding impulse buys, and becoming indispensable through devoted, excellent work, with Michelangelo, Martin Luther King and Lincoln as examples. Real wealth also includes things money can't buy.

## Week 7: Health and Love

Health affects personality, first impressions and employability, since healthy people are seen as disciplined and reliable. Managing it rests on:

- Regular exercise (morning is best, with tips for sticking to it).
- Diet: regular meals, a heavy breakfast and a light early dinner, fruits and vegetables, about 2 litres of water, and no white foods or soft drinks.
- Sleep: 7-8 hours, a good sleep environment, and avoiding gadgets, late heavy meals and alarms.

Love is defined broadly, drawing on Shakespeare, D. H. Lawrence, Gibran, Lovelace and Corinthians: real love is constant, free, patient and unconditional. Managing love means reading body language and timing, letting relationships mature through stages, handling rejection gracefully and keeping romance separate from work.

## Week 8: Ethics, Etiquette, Mind, Memory and Environment

Values, ethics and etiquette make up character (“when character is lost, everything is lost”). Professional ethics combine written codes with society’s moral expectations. Poor etiquette and business etiquette are covered in detail: introductions, not interrupting, confidentiality, not criticising your company, and praising in public but reprimanding in private.

The mind is powerful but scattered, and it can be trained through auto-suggestion, positive self-talk and effective thinking (gather facts, sleep on it, trust your gut). Memory techniques include association through absurd images, remembering names and faces, number-image links, memory aids and mental exercise. The final lecture urges care for the environment, with practical steps (save water and electricity, avoid plastic, use public transport, plant trees) and a shift from anthropocentric to eco-centric thinking. It closes with Gandhi’s line that the earth provides enough for everyone’s need but not everyone’s greed.

## SWOC Presentation

Strategic Career Roadmap: SWOC Analysis (slides)

# Strategic Career Roadmap: SWOC Analysis

Path to Principal Cloud Architect / Director of Engineering

**Candidate:** Inba Ilakiyan | 1st Year B.E./B.Tech CSE | Batch 2026 – 2030

*Goal: get deep into cloud and systems early, then move toward architecture and engineering leadership*

## Executive Summary & Career Vision

**1 Vision**

Start as a first-year CSE student and grow into a Principal Cloud Architect. Director of Engineering is the leadership route.

**2 Objective**

Learn Linux and networking properly, then cloud. Put design, judgment and problem-solving ahead of routine coding.

**3 Framework**

Use a SWOC matrix (Strengths, Weaknesses, Opportunities, Challenges) to draw up a 4-year milestone roadmap.

# SWOC Matrix at a Glance

## Strengths

Internal

Excellent problem solver and Linux enthusiast, with Python projects on GitHub

## Weaknesses

Internal

Impatient and quick to rush in; beginner in systems; no certifications yet

## Opportunities

External

AI lifts the coding baseline, which pushes value toward architecture and design

## Challenges

External

Noisy, overstimulating environments; a shrinking junior pipeline

# Strengths

| | | |
|---|---|---|
| 1 | **Excellent problem solver** | I can sit down and think a complex problem through. That is the core skill of architecture. |
| 2 | **Linux enthusiast** | I'm comfortable working terminal-first, and I like finding out how systems work. |
| 3 | **Python foundation** | Hands-on practice through two personal projects, Royal-Roll and latte. |
| 4 | **Public portfolio started** | An active GitHub profile with a README, so recruiters can see the projects. |

# Weaknesses

### Working style

Impatience: I tend to rush into a problem and hit a brick wall before I've planned anything.

It's the habit most worth fixing. Architects earn their keep by thinking before they build.

### Experience gaps

I'm a beginner in operating systems, networks and databases, none of which I've studied formally yet.

No production experience, internship or certification so far.

# Opportunities

### AI raises the coding baseline

More and more routine boilerplate gets automated, so value moves toward architecture, systems design and decision-making.

### Growing cloud demand

Migration, DevOps, platform engineering and FinOps roles keep growing.

### Student programs

AWS Educate, the GitHub Student Developer Pack and Azure for Students all offer free credits.

### Certifications and community

Entry-level certs such as AWS Cloud Practitioner and AZ-900, plus open source, hackathons and meetups.

# Challenges

### Overstimulating environments

Noisy, high-energy settings overstimulate me. That hits labs, hackathons and open-plan offices.

### Shrinking junior pipeline

That same AI shift means fewer boilerplate junior roles, so I need to show design ability early.

### Experience paradox

Entry roles often ask for experience I haven't had a chance to get yet.

### Tooling churn and cloud cost

Tools change fast, and personal labs risk surprise bills.

# Action Plan: Closing the Gaps

### 1 Impatience

Before any project, I write a short design note (goal, components, risks) and sketch a diagram. Only then do I build.

### 2 Knowledge gap

Use Linux every day. Study networking basics (TCP/IP, DNS, subnetting) alongside coursework.

### 3 Credential gap

Take AWS Cloud Practitioner before entering Year 2.

# Action Plan: Using Opportunities

### Use AI as a tool

Let it write the boilerplate. I focus on design, review and debugging, and I should be able to explain every line I ship.

### Manage noise

Book deep-work blocks for quiet hours, wear headphones, and choose async or small-group formats where possible.

### Build production-style projects

CI/CD, IaC and containerized apps, each with an architecture diagram in its README.

### Stay cost-safe

Set billing alarms. Run every cloud lab on student credits.

# 4-Year Milestones Matrix

| Year | Primary Focus | Target Deliverable | Target Credential |
|---|---|---|---|
| **Year 1 (2026–27)** | Linux, networking, cloud basics | Portfolio site on AWS S3 / CloudFront | AWS Cloud Practitioner |
| **Year 2 (2027–28)** | Docker, Python automation, IaC | 2-tier VPC deployed with Terraform | AWS Solutions Architect Associate |
| **Year 3 (2028–29)** | Kubernetes, CI/CD, DevSecOps | Microservices on EKS via GitHub Actions | CKA or Terraform Associate |
| **Year 4 (2029–30)** | SRE, FinOps, leadership | Multi-cloud disaster recovery capstone; lead a team project | Placement / off-campus hiring |

# Conclusion & Key Takeaways

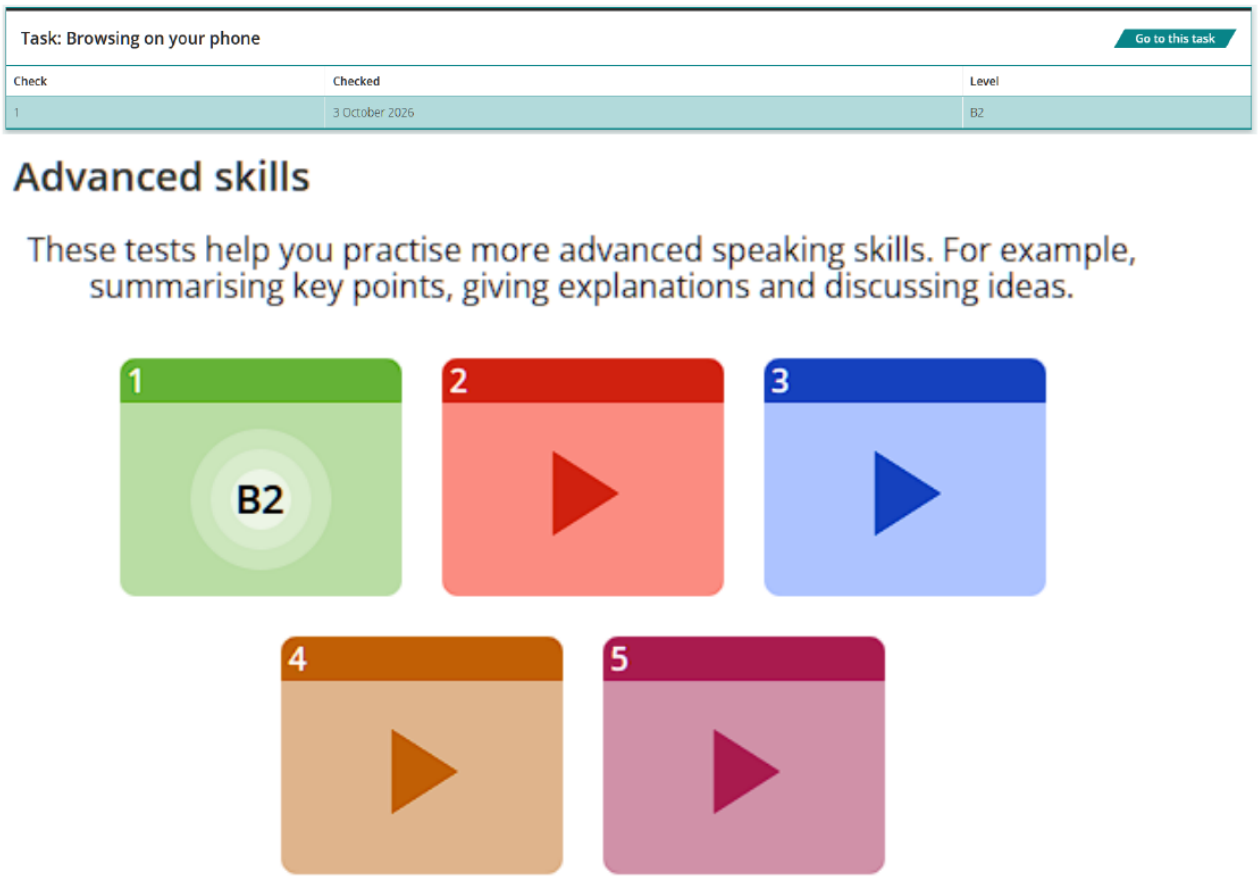
1. **Think first, then build**
   Planning before coding is what turns my problem-solving strength into architect-level work.

2. **Fundamentals plus AI**
   AI-generated code is only trustworthy when someone understands the systems underneath it.

3. **Verdict**
   If I plan with discipline and keep doing hands-on cloud work, I should be able to build a profile aimed at architecture and engineering leadership.

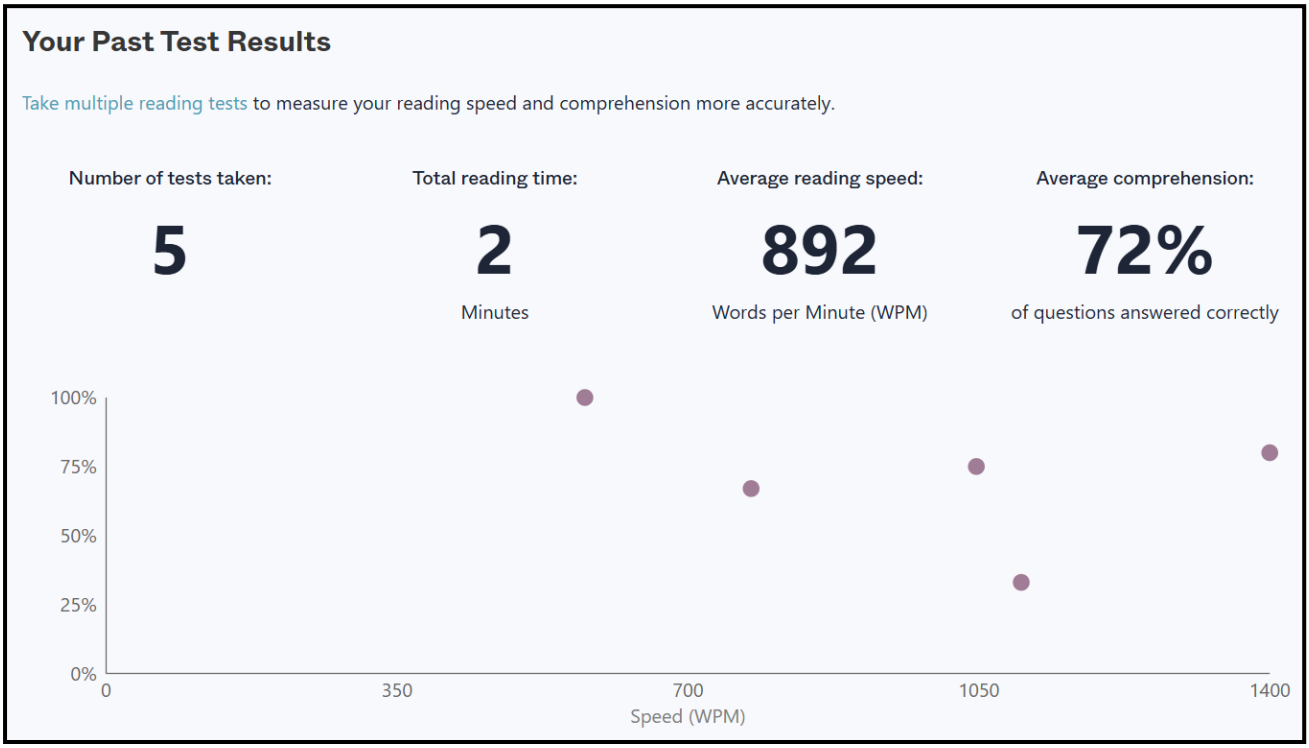
# Write and Improve / Speak and Improve

Speak and Improve: score sheet (Task: Browsing on your phone)

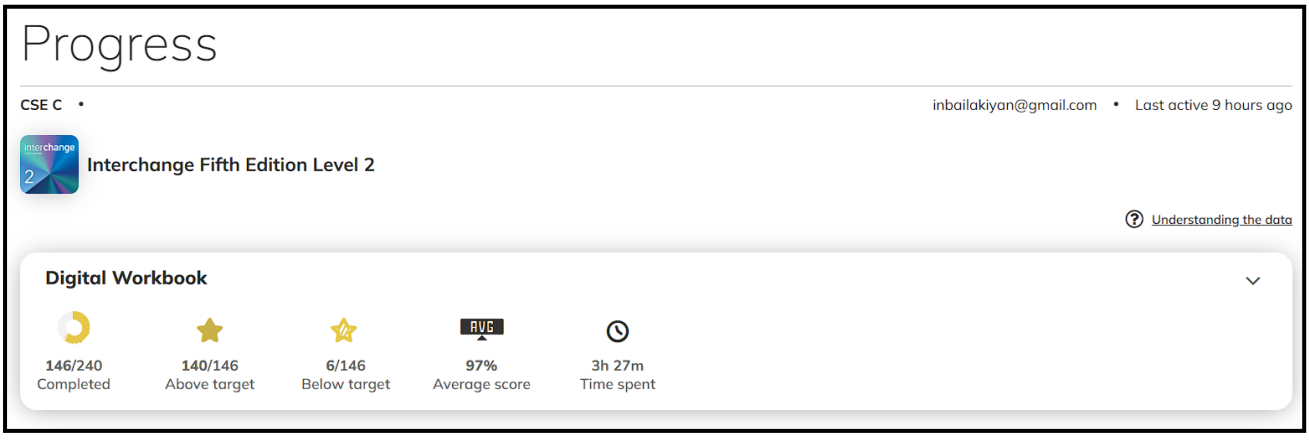


# Write and Improve / Speak and Improve

Reading speed test results and Interchange 2 progress (screenshots)



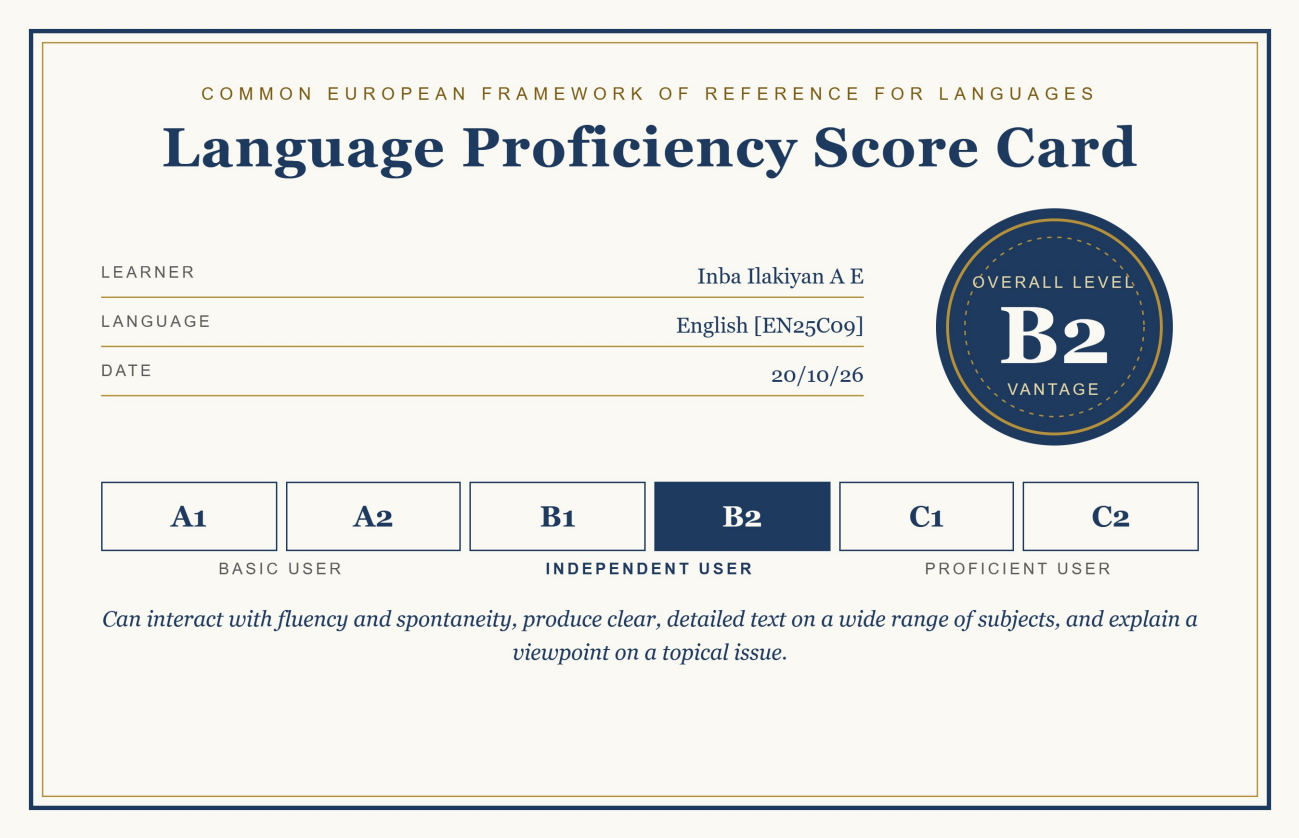

**Reading speed test: 5 tests, average 892 WPM, 72% comprehension.**




**Interchange Fifth Edition Level 2: Digital Workbook progress.**

# Write and Improve / Speak and Improve

CEFR score card



COMMON EUROPEAN FRAMEWORK OF REFERENCE FOR LANGUAGES

## Language Proficiency Score Card

LEARNER: Inba Ilakiyan A E

LANGUAGE: English [EN25C09]

DATE: 20/10/26

OVERALL LEVEL B2 VANTAGE

| A1 | A2 | B1 | B2 | C1 | C2 |
|---|---|---|---|---|---|
| BASIC USER | | INDEPENDENT USER | | PROFICIENT USER | |

*Can interact with fluency and spontaneity, produce clear, detailed text on a wide range of subjects, and explain a viewpoint on a topical issue.*

**My CEFR level is B2 (Upper Intermediate).**

# Professional English Practice Worksheets – Solved

*Course Code: EN25C09 | Subject: Professional English – I (Lab Integrated) | Worksheet 1: Model Exam I (Parts A, B, C), followed by Worksheet 2: Model – I Practice Worksheet.*

## Worksheet 1 (EN25C09 Model Exam I) – Part A (15 × 1 = 15 Marks)

1. **Answer:** monitors
2. **Answer:** recorded
3. **Answer:** stronger
4. **Answer:** must
5. **Answer:** can
6. **Answer:** The laboratory developed a new testing device that measures material strength with high precision.
7. **Answer:** rises
8. **Answer:** would not have failed
9. **Answer:** The mechanics installed dual damper springs to isolate chassis resonance.
10. **Answer:** optimizing
11. **Answer:** 50 mL of hydrochloric acid was poured into the titration flask.
12. **Answer:** smoothly
13. **Answer:** The project supervisor asked the engineer whether he had calibrated the pressure sensor before testing.
14. **Answer:** The incorrect noun form is "equipments", because "equipment" is uncountable. Correct sentence: The laboratory purchased several new pieces of equipment for conducting material testing.
15. **Answer:** worse

## Worksheet 1 – Part B (10 × 2 = 20 Marks)

1. **Answer:** The engineer calibrated the pressure sensor and verified its accuracy.
2. **Answer (Portability):** A laptop is more portable than a desktop computer because it is lighter and can be carried anywhere.

   **Answer (Screen Size):** A desktop computer usually has a larger screen than a laptop, so it is more comfortable for long hours of detailed work.
3. **Answer:** The engineer designed a new cooling system, which reduces the temperature of industrial machines.
4. (a) **Answer:** If the battery voltage drops below the required level, the device will shut down automatically.

   (b) **Answer:** If the technician had checked the connections earlier, the fault would have been detected before the breakdown.
5. **Answer:** The solution was heated to 80°C before testing.
6. **Answer:** The project manager asked the technician whether he had inspected the pressure valves before starting the compressor.
7. (a) **Answer:** The engineer received a lot of feedback on the new design. ("Feedback" is uncountable, so it cannot take "many" or the plural form "feedbacks".)

   (b) **Answer:** The students received several pieces of useful advice from their teacher. ("Advice" is uncountable, so "advices" is wrong.)
8. (a) **Answer:** The technician wears safety gloves to protect his hands.

   (b) **Answer:** The cooling fan operates continuously to prevent the motor from overheating.
9. (a) **Answer:** efficiently

   (b) **Answer:** carefully

10. (a) **Answer:** can

    (b) **Answer:** must not

## Worksheet 1 – Part C (5 × 13 = 65 Marks)

### Question 26 (a) Safety instructions for safe and responsible use of the internet

1. Use a strong, unique password of at least twelve characters, mixing letters, numbers and symbols, for every account.
2. Enable two-factor authentication on your email, banking and social media accounts.
3. Do not click on links or open attachments from unknown or suspicious senders.
4. Never share personal details such as OTPs, passwords, Aadhaar or bank details with anyone online.
5. Check that a website address begins with "https" and shows a padlock before entering any sensitive information.
6. Keep your operating system, browser and antivirus software updated at all times.
7. Avoid banking or shopping on public Wi-Fi; use a trusted VPN if you must connect.
8. Report cyberbullying, online fraud and phishing attempts immediately to a trusted adult, the platform, or the National Cyber Crime Helpline (1930).

### Question 27 (a) Descriptive essay: A traditional art form in my region and how modern initiatives can preserve it

**Selected Question: 27.a**

**Bharatanatyam: The Dance That Carries Tamil Nadu's Soul**

Among the many traditional art forms of Tamil Nadu, Bharatanatyam is the one that has carried our culture across the centuries. It began inside the temples, where devadasis performed it as an offering to the deity. The epic Silappathikaram describes classical dance in detail, and the carvings on the Brihadeeswarar Temple at Thanjavur and the Chidambaram Nataraja Temple show its postures in stone. In the nineteenth century, the Thanjavur Quartet organised the repertoire into the form we see today.

Traditionally, a dancer trains for many years under a guru. A performance blends nritta (pure rhythm), nritya (expressive storytelling) and natya (drama), told through hand gestures called mudras, facial expressions and Carnatic music. The dancer wears a bright silk costume, temple jewellery and ankle bells called salangai. The first full solo performance, the arangetram, marks the end of a student's formal training.

Today, many young people feel the art form is old-fashioned and too expensive to learn. Modern initiatives can change this. Schools and colleges can include classical dance in cultural clubs and the curriculum, as institutions like Kalakshetra in Chennai already do. Online classes, recorded performances and social media reels can reach learners in small towns. Festivals such as the Margazhi season in Chennai give young artists a stage, and scholarships can help families who cannot afford training.

Bharatanatyam is more than entertainment; it is a living record of our history, music and devotion. If we support it through education, technology and public platforms, the next generation will carry the dancing bells forward.

### Question 28 (b) Formal letter of request to the Principal

**Selected Question: 28.b**

Inba Ilakiyan
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Principal
St. Joseph's College of Engineering

OMR, Chennai – 600 119

**Subject: Request for permission to attend a technical workshop on Artificial Intelligence and Emerging Technologies**

Respected Sir/Madam,

I am a first-year B.E. Computer Science and Engineering student of your college. I have come to know that [Name of Institution] is organising a one-day technical workshop on Artificial Intelligence and Emerging Technologies on [date]. I request your kind permission to attend this workshop.

My objective in attending is to learn how artificial intelligence is applied in real engineering problems and to gain hands-on exposure to tools that are not covered in detail in our first-year syllabus. I also hope to meet experts and students from other institutions and learn from their experience.

I expect to learn the basics of machine learning, practical uses of AI in software and cloud systems, and the latest trends in emerging technologies. After the workshop, I will share the notes and key points with my classmates and use what I learn in my Python projects.

I request you to kindly grant me on-duty leave for the day of the workshop and permit me to travel to the venue. I will make up for any classes missed and will submit the participation certificate to the department after the workshop.

I would be grateful if you could grant me this permission.

Thank you for your kind consideration.

Yours obediently,
Inba Ilakiyan
(Register No.: [Register No.])

## Question 29 (a) Analytical essay: AI-based assessment versus traditional teacher-based assessment

**Selected Question: 29.a**

**AI-Based Assessment versus Traditional Teacher-Based Assessment in Engineering Education**

Assessment decides how students learn, so the choice between AI-based and teacher-based methods matters in engineering education. Each has clear strengths and weaknesses.

In terms of accuracy, AI tools are consistent. An automated system can evaluate hundreds of code submissions or objective answers using the same rules every time, without getting tired. Teachers, on the other hand, can understand a student's reasoning in a long derivation or a design report, which software often misses. Therefore, AI is accurate for objective answers, whereas teachers are better for open-ended ones.

Feedback time is where AI wins most clearly. Students receive results and error explanations within seconds, while manual evaluation of a large class may take weeks. Quick feedback lets students correct their mistakes while the topic is still fresh. Conversely, a teacher's comments arrive more slowly but are usually more personal and motivating.

Regarding fairness, AI does not show favouritism or mood-based marking. However, it can carry hidden bias from the data it was trained on, and it may penalise students who give unusual but valid solutions. A teacher can notice such cases and adjust the marks.

Finally, human judgement remains important. Engineering involves creativity, ethics and teamwork, which cannot be reduced to a score.

In conclusion, neither method is perfect alone. The best approach is a blend, where AI handles routine and objective checking quickly, and teachers review complex work and guide students. This combination keeps assessment fast, fair and meaningful.

## Question 30 (b) Personal email: Orientation programme

**Selected Question: 30.b**

**To:** [friend's email address]

**Subject:** My first-year orientation programme at college

Dear [Friend's name],

How are you? I hope you are doing well and enjoying your college life too. I have been wanting to write to you about my orientation programme, so here it is.

The programme was held for the first-year engineering students at our college. It began with a welcome address by the Principal, followed by talks from senior faculty and some alumni. We also had fun icebreaker games and team activities, which helped me make new friends within the first day itself.

The best part was the tour of the academic facilities. We visited the library, the computer labs, the workshops and the seminar halls, and the staff explained how to use each of them. We were also told about the college rules on attendance, discipline, dress code and examinations, so now I know exactly what is expected from us.

Overall, the programme made me feel much more comfortable in the new environment. I understood how college life works and how to plan my studies, and I felt ready to start my academic journey with confidence.

Write back soon and tell me about your college too. Take care!

Your friend,
Inba

## Worksheet 2 (Model – I Practice Worksheet) – Exercises 1 to 15

### Exercise 1: Simple Present Tense (Subject-Verb Concord)

1. recognize
2. requires
3. transmit
4. agrees
5. form

### Exercise 2: Simple Past Tense

1. conducted
2. deployed
3. opened
4. published
5. detected

### Exercise 3: Comparative Adjectives

1. higher
2. safer
3. faster
4. more durable
5. lower

### Exercise 4: Modals of Obligation and Necessity

1. must
2. must
3. have to
4. must
5. has to

### Exercise 5: Modals of Capability and Possibility

1. can
2. may
3. can
4. can
5. can

### Exercise 6: Relative Clauses

1. The civil engineer who designed the suspension cable bridge received the national infrastructure award.
2. The spectrometer, which analyzes infrared emission spectra, is mounted inside the cleanroom vacuum chamber.
3. The technical specialist whose patent on pneumatic grippers was approved was reassigned to the robotics lab.
4. This microcontroller, which features ultra-low power sleep states, is ideal for battery-operated remote sensors.
5. The field researchers who surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

### Exercise 7: If-Clauses (Type 1)

1. exceeds
2. fails
3. engages
4. rise
5. registers

### Exercise 8: If-Clauses (Type 3)

1. had started
2. had lubricated
3. would have been prevented
4. would have tripped
5. had remained

### Exercise 9: Infinitives of Purpose

1. The mechanics installed dual damper springs to isolate chassis resonance.
2. Technicians purge the fuel line with dry nitrogen gas to remove residual flammable vapors.
3. We introduced an optical encoder to determine spindle velocity accurately.
4. The developers wrote an automated load testing script to simulate ten thousand concurrent connections.
5. Gold plating is applied to terminal connectors to prevent atmospheric corrosion.

### Exercise 10: Gerunds after Prepositions

1. redesigning
2. replacing
3. reducing
4. cleaning
5. filtering

### Exercise 11: Impersonal Passive Voice

1. 50 mL of hydrochloric acid was poured into the titration flask.
2. The robotic arm was tested under a 50 kg dynamic payload.
3. The optical pyrometer was calibrated prior to ignition.
4. Severe cavitation damage was observed along the impeller blades.
5. Volatile solvents are stored in flame-resistant containment cabinets.

### Exercise 12: Adverbs of Manner

1. precisely
2. periodically
3. rapidly
4. firmly
5. smoothly

### Exercise 13: Reported Speech

1. The quality manager asked the technician whether the insulation resistance test had been documented.
2. The lead architect enquired whether the server cluster could handle double the transaction throughput.
3. The field auditor asked the safety engineer whether he had inspected the gas leakage relief lines.
4. The client asked the team whether the firmware update would fix the intermittent battery drain.
5. The instructor asked the lab group why the multimeter had shown fluctuating voltage readings.

### Exercise 14: Countable and Uncountable Nouns

1. Incorrect: "machineries" and "equipments". Correct: The procurement officer purchased several new pieces of machinery and equipment for the workshop.
2. Incorrect: "datas". Correct: The laboratory generated a huge amount of test data during the wind tunnel trials.
3. Incorrect: "softwares". Correct: We need three more software licenses to complete the finite element analysis on the model.
4. Incorrect: "informations". Correct: The field report provided a lot of valuable information regarding the soil composition.
5. Incorrect: "apparatuses" and "glasswares". Correct: All chemical apparatus and glassware must be washed thoroughly after the experiment.

### Exercise 15: Comparative Adjectives (Irregular Forms)

1. worse
2. better
3. farther
4. less
5. further

## Worksheet 2 – Part B (Exercises 1 to 10)

### Exercise 1: Simple Past Tense Conversion

1. The technician inspected the solar arrays and recorded the peak power conversion value.
2. The chemical analyzer detected trace mercury levels and sent an automated alert to the environmental dashboard.
3. Maintenance crews lubricated the heavy conveyor bearings and replaced the worn rubber belts before the morning shift.

4. The supervisory software polled field sensors every five milliseconds and archived temperature trends into a secure database.
5. The testing team applied a 500-volt dielectric pulse across the stator winding and monitored for insulation breakdown.

### Exercise 2: Comparative Sentences

1. Cloud storage has a lower initial deployment cost than on-premise physical servers because no hardware has to be bought. It is also more elastic, since its capacity can be increased within minutes.
2. Fiber-optic cables offer a much higher data transfer bandwidth than copper twisted-pair cables. They are also less susceptible to electromagnetic interference than copper cables.
3. Electric vehicles produce lower tailpipe emissions than internal combustion engine vehicles, as they release none at all. However, EVs generally have a higher initial purchase price than ICE vehicles.
4. Hydraulic actuators produce a greater raw force output than electromechanical actuators. They also carry a higher risk of fluid leakage, because electromechanical actuators use no fluid.
5. Relational (SQL) databases have a more rigid schema than document (NoSQL) databases. Document databases usually provide higher horizontal query throughput than relational databases.

### Exercise 3: Relative Clauses

1. The algorithm that analyzes microscopic cracks in turbine blades was patented by the aeronautics team.
2. The lead chemical engineer, who noticed an abnormal pressure spike in chamber two, authorized the emergency boiler shutdown.
3. The laboratory houses a 400 kV impulse voltage generator, where high-voltage insulator puncture tests are conducted.
4. We engaged an external cybersecurity consultant whose firmware vulnerability audit exposed three backdoor ports.
5. The supervisory system, which utilizes an artificial neural network, predicts transformer insulation life based on dissolved gas concentrations.

### Exercise 4: If-Clauses

1. (a) If the operating temperature exceeds 85°C, the cooling system will switch on automatically.

   (b) If the operator had pressed the emergency shutdown switch immediately, the accident would have been avoided.
2. (a) If the supply voltage drops below the 200V threshold, the backup power supply will take over.

   (b) If the network administrator had enabled multi-factor authentication, the unauthorized access would have been prevented.
3. (a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, the ventilation fans will start and an alarm will sound.

   (b) If the mechanical team had checked the bearing lubrication schedule, the bearing would not have overheated.
4. (a) If the autonomous guided vehicle (AGV) detects an obstacle within 1 meter, it will stop immediately.

   (b) If the thermal sensor had registered the runaway reaction in time, the explosion would have been prevented.
5. (a) If the software receives an unauthenticated API token, it will reject the request and return an error message.

   (b) If the civil inspectors had reviewed the core sample density tests, they would have detected the weak foundation before construction.

### Exercise 5: Impersonal Passive Voice

1. The crude extract was distilled at a constant temperature of 60°C.

2. The cross-section of the weld seam was polished with fine diamond paste.
3. Fifteen cast-iron pump casings were rejected due to internal porosities.
4. The resistance across the bridge circuit was measured using a calibrated digital ohmmeter.
5. The emergency aerodynamic braking flaps were initiated at an altitude of 3,000 meters.

### Exercise 6: Reported Speech

1. The safety supervisor asked the intern whether he had verified the grounding connections on the generator.
2. The plant director asked the site engineer whether the emergency effluent treatment plant had received clearance.
3. The IT security lead enquired of the developer why he had disabled the firewall rule on the database server.
4. The client asked the project manager whether the bridge structure could withstand Category 4 hurricane winds.
5. The research guide asked the postgraduate scholar how many sample trials he had conducted under cryogenic conditions.

### Exercise 7: Countable vs. Uncountable Nouns

1. (a) We need to install several new pieces of software and machinery in the CAD laboratory.
   (b) The team collected plenty of useful information from the field test.
2. (a) The technician ordered ten new pieces of apparatus and modern electronic equipment for the antenna lab.
   (b) High-speed train operations require advanced transport infrastructure and dedicated electrical power.
3. (a) The environmental sensor gathers extensive atmospheric data and pollution indices every minute.
   (b) The construction crew unloaded tons of sand, cement, and steel debris at the worksite.
4. (a) All electrical hardware and wiring must comply with local safety legislation.
   (b) The data scientist discovered several interesting insights after running the machine learning clustering.
5. (a) Laboratory staff must clear away all glassware and acidic waste before leaving the premises.
   (b) The project failed due to a lack of proper funding and technical research.

### Exercise 8: Infinitives of Purpose and Gerund Constructions

1. (a) A pressure relief valve is installed to vent excess boiler steam.
   (b) The automated script runs every midnight to archive legacy server logs.
2. (a) Automotive engineers installed an intercooler to reduce the compressed air intake temperature.
   (b) The technicians analyzed the structural stress points by placing piezoelectric strain gauges along the girder.
3. (a) High-performance heatsinks feature copper heatpipes to conduct heat away from the silicon die.
   (b) The factory cut water consumption by 30% by implementing a closed-loop recycling filtration system.
4. (a) Safety guards are mounted over rotating flywheel belts to prevent accidental worker contact.
   (b) The algorithm optimizes memory bandwidth by pre-fetching sequential array indices into L2 cache.
5. (a) The electronics lab installed an uninterruptible power supply (UPS) to protect servers against sudden voltage sags.
   (b) The metallurgy group enhanced surface hardness by diffusing nitrogen atoms into the steel casing.

### Exercise 9: Adverbs of Manner

1. (a) automatically
   (b) flawlessly
2. (a) accurately

(b) instantly

3. (a) continuously
   (b) cautiously
4. (a) consistently
   (b) catastrophically
5. (a) periodically
   (b) smoothly

### Exercise 10: Modals

1. (a) can
   (b) must not
2. (a) must
   (b) may
3. (a) must not
   (b) could
4. (a) should
   (b) must
5. (a) can
   (b) must not

## Worksheet 2 – Topic 1: Writing Imperative Safety Instructions

### Question 1: Chemistry and Materials Testing Laboratory

1. Wear a lab coat, safety goggles and nitrile gloves at all times inside the laboratory.
2. Handle concentrated acids and alkalis only inside the fume hood.
3. Add acid to water slowly; never add water to acid.
4. Do not smell, taste or touch any chemical directly.
5. Label every container clearly, and never return unused chemicals to the stock bottle.
6. Clean all glassware and testing equipment after each experiment to avoid contamination.
7. Keep food, drinks and open flames away from the work area.
8. Report every spill, burn or breakage to the instructor at once, and use the eyewash station or safety shower in an emergency.

### Question 2: High-Voltage Electrical Machines and Drives Laboratory

1. Make sure the main supply is switched off before making or changing any connection.
2. Wear insulated rubber gloves and safety shoes, and stand on an insulated rubber mat near live panels.
3. Never touch exposed terminals, bare wires or rotating parts while the machine is running.
4. Get the circuit connections checked by the instructor before switching on the supply.
5. Know the location of the circuit breakers, fuses and emergency stop button, and keep them in working condition.
6. Remove rings, watches and metal jewellery, and avoid loose clothing, to prevent arc flash burns and accidental short circuits.
7. Keep water and wet hands away from electrical equipment, and use only insulated tools.
8. Increase the motor load gradually, and switch off the supply and discharge the capacitors before disconnecting the circuit.

### Question 3: Advanced Manufacturing and CNC Machining Workshop

1. Wear safety glasses, safety shoes and close-fitting clothes, and tie back long hair.
2. Keep the machine door and guard closed while the spindle is running.
3. Do not touch the spindle, cutter or workpiece until the machine has stopped completely.
4. Clamp the workpiece and tool holder firmly before starting the program.
5. Use a brush or chip hook, and heat-resistant gloves, to remove hot metal chips, and only after stopping the machine.
6. Wipe up coolant spills immediately, and keep the floor dry to prevent slipping.
7. Run the program in simulation mode first, and keep your hand near the emergency stop button during the first cut.
8. Never leave the machine unattended or operate it without the instructor's permission.

## Worksheet 2 – Topic 2: Free Writing – Festivals and Cultural Celebrations

### Question 1: A traditional harvest festival and eco-friendly practices

**Pongal: The Harvest Festival of Tamil Nadu and How Eco-Friendly Practices Can Protect Its Spirit**

Pongal is the most important harvest festival of Tamil Nadu. It is celebrated in mid-January to thank the Sun God, the rain and the cattle for a good harvest. The festival lasts four days: Bhogi, Thai Pongal, Mattu Pongal and Kaanum Pongal.

On Bhogi, families clean their homes and throw away old things. On Thai Pongal, they draw colourful kolams at the entrance and cook fresh rice and milk in a new clay pot in the open yard. When the milk boils over, everyone shouts "Pongalo Pongal!" The pot is decorated with turmeric plants and sugarcane. The festive meal includes sakkarai pongal, venn pongal, sambar, vadai and payasam, served on banana leaves. On Mattu Pongal, farmers decorate and worship their cattle, and Kaanum Pongal is a day for family visits and community games.

Modern eco-friendly practices can preserve the core spirit of the festival. Instead of burning plastic and old tyres on Bhogi, communities can burn only dry leaves or turn them into compost, which reduces smoke and air pollution. Clay pots, banana leaves and areca-leaf plates can replace plastic and thermocol, and kolams can be drawn with natural rice flour. Villages can organise tree-planting drives, rainwater harvesting and cloth-bag campaigns during the festival, and use solar lamps instead of excess electric decorations.

In this way, Pongal can remain joyful and traditional while staying kind to nature, and the next generation will learn that celebration and conservation can go together.

### Question 2: A temple car festival and sustainable measures

**The Temple Car Festival: Faith, Food and a Cleaner Celebration**

In my hometown, as in many towns of Tamil Nadu, the temple car festival, or Ther Thiruvizha, is the biggest event of the year. Thousands of devotees gather in the streets to watch the deity pulled in a huge, richly decorated wooden chariot, called the ther.

The festival begins with flag hoisting and several days of processions. On the car day, devotees of all ages hold thick ropes and pull the heavy chariot together through the four main streets, chanting prayers while drums and nadaswaram fill the air. This shared effort is a strong symbol of unity, because people from every community take part. Women draw kolams on the road, and families offer coconuts, bananas and flowers to the deity. Volunteers give free food such as pongal, puliyodharai, curd rice and sweet payasam as prasadam, served on leaves.

However, such a large crowd also creates problems. Single-use plastic cups, plates and bags litter the streets, and loudspeakers cause noise pollution. Eco-friendly measures can solve these problems. Organisers can ban single-use plastic and provide steel tumblers, banana leaves and paper bags. Dustbins and waste-sorting points can be placed along the route, and volunteers can collect and compost the food waste. Noise can be controlled by limiting loudspeaker volume and using traditional instruments. Barricades, marked entry and exit lanes, volunteers, and first-aid points can manage the crowd safely.

With these steps, the festival can keep its sacred joy and remain clean and peaceful.

### Question 3: Regional folk art and handicraft fairs with renewable energy and zero-waste practices

**Folk Art and Handicraft Fairs: Heritage with a Green Future**

A regional folk art and handicraft fair is a colourful meeting of tradition and community. Walking through the fair grounds, visitors see Karagattam and Oyilattam performances, hear the beat of the parai drum, and watch artisans make Thanjavur dolls, palm-leaf baskets, clay pots and silk sarees. The air smells of fresh filter coffee, murukku, adhirasam and hot sundal.

These fairs are important because they give artisans a market and help young people see the value of their heritage. Many stalls offer live demonstrations, so children can try pottery or weaving themselves. Local delicacies prepared by village cooks add to the festive atmosphere and support small businesses.

However, large fairs use a lot of electricity and produce a lot of waste. Modern environmental strategies can solve this. Solar micro-grids, made of rooftop panels and batteries, can power the lights, sound systems and stalls without diesel generators. Zero-waste composting can turn leftover food and leaves into manure for local farms. Organisers can use reusable steel plates, clay cups and cloth banners, provide separate bins for wet and dry waste, and recycle paper and cardboard.

If we combine the traditional skill of our artisans with clean energy and responsible waste management, these fairs can protect our culture and our planet together.

## Worksheet 2 – Topic 3: Free Writing – Food and Technical Travel Plans

### Question 1: Technical study tour to a major hydroelectric dam

**A Technical Study Tour to the Mettur Hydroelectric Dam**

Our department has planned a one-day technical study tour to the Mettur Dam on the Cauvery River in Salem district. The group of forty students and two faculty members will leave the college by a chartered bus at 4:30 a.m. and travel along the national highway, with a short breakfast halt on the way. The journey takes about six hours, and we will return by night. Students will carry ID cards, water bottles and a notebook.

After registering at the gate, an engineer will take us through the powerhouse. The long concrete dam holds back the huge Stanley Reservoir. Inside the powerhouse, we will see the penstocks, the turbines and the generators, and learn how falling water turns turbines to produce electricity. We will also observe the control room, where operators monitor water level, flow and power output on screens, and the spillway gates used to release extra water safely.

For lunch, we will try local dishes such as hot rice with sambar, fish fry from the river and fresh mango juice, which make the trip a cultural experience as well. In the evening, we will walk through the nearby park and enjoy the green landscape and the calm reservoir.

The visit will help us understand renewable energy, water management and the role of civil, mechanical and electrical engineering in one project. We will follow all safety rules, wear helmets where needed, and stay with our group at all times.

### Question 2: Technical study tour to an integrated solar and wind hybrid energy farm

**A Technical Study Tour to a Solar and Wind Hybrid Energy Farm**

We plan a two-day study tour to the Muppandal wind farm region near Kanyakumari, where wind turbines and solar panels work together. Thirty students and two faculty members will take the overnight train from Chennai to Nagercoil, then travel about one hour by a hired van to the farm. We will stay one night in a budget lodge nearby, and return by train on the second night.

On arrival, a site engineer will give a short briefing on how the hybrid plant works. The wind turbines turn on tall towers, and their blades rotate a generator inside the nacelle. We will watch how the strong coastal wind drives them, and how the blades adjust their angle with the wind speed. In the solar section, we will observe rows of photovoltaic panels, inverters and transformers that feed power into the grid. The control room will show how the two sources balance each other: solar power peaks in the afternoon, whereas wind is strong in the evening and

night.

For food, we will taste Nanjil Nadu specialities such as fish curry with rice, banana chips, avial and fresh coconut sweets. Eating in small local hotels along the route will also help us understand the region's culture.

The tour will show us the practical side of renewable energy, and we will follow safety rules by wearing helmets, keeping away from the turbine bases and obeying the engineers' instructions.

### Question 3: Technical field visit to an automated seawater reverse osmosis desalination plant

**A Technical Field Visit to a Seawater Reverse Osmosis Desalination Plant**

Our class is planning a half-day field visit to a seawater reverse osmosis desalination plant on the East Coast Road near Chennai, such as the Nemmeli plant. About fifty students and three faculty members will travel in two college buses, leaving at 8:00 a.m. and reaching the plant in about ninety minutes. We will carry ID cards and wear closed shoes for safety.

At the plant, the engineers will first explain the process. We will see the intake pumping station, which draws seawater through long pipes under the sea bed. The water then passes through pre-treatment filters that remove sand and suspended particles. Next, high-pressure pumps push the water through the membrane filtration racks, where the salt is separated and fresh water is produced. We will also see the brine discharge outlet, where the concentrated salt water is carefully diluted and released back into the sea to protect marine life. The control room shows the whole process through automated screens.

After the visit, we will have lunch at a local seaside restaurant. The coastal menu includes fish fry, prawn masala, crab curry with steamed rice and filter coffee, which gives us a taste of the Chennai coastal culture.

The visit will help us see how engineering solves water scarcity, how energy is used in the process, and why environmental care in brine disposal is important.

## Worksheet 2 – Topic 4: Formal Letters of Request

### Question 1: Letter to the Head of the Computer Science Department (lab access for a hackathon)

Inba Ilakiyan
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Head of the Department
Department of Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

**Subject: Request for permission and lab access after college hours to prepare for a National AI Hackathon**

Respected Sir/Madam,

I am writing on behalf of my project team of four first-year students. We have registered for a National AI Hackathon to be held on [date], and we respectfully request permission to use the computer laboratory after college hours.

The objective of our project is to build a simple AI-based solution that solves a real campus problem, using Python and machine learning. We need about three weeks to design, train and test the model, and our home systems are not powerful enough for this.

We require access to the lab between 4:30 p.m. and 7:30 p.m. on weekdays, with Python, Jupyter Notebook, TensorFlow or scikit-learn, Git and a stable internet connection. We promise to follow all lab rules, maintain the equipment carefully, and leave the lab clean and switched off after use. We will also work under the supervision of a faculty member if required.

We would be very grateful if you could grant us this permission.

Thank you for your kind consideration.

Yours obediently,
Inba Ilakiyan
(On behalf of the project team)

### Question 2: Letter to the Plant Manager (industrial plant visit)

Inba Ilakiyan
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Plant Manager
[Name of Factory], Industrial Robotics and Automation Division
[Address]

**Subject: Request for permission to conduct an industrial visit for 50 first-year engineering students**

Respected Sir/Madam,

I am a first-year engineering student of St. Joseph's College of Engineering, Chennai. On behalf of my class, I request your permission to visit your robotics and automation factory with fifty students and two faculty members on a convenient date in [month].

The intended learning outcomes of the visit are to understand how robots and automated systems work on a real production line, to see how sensors, controllers and robotic arms are combined, and to learn about industrial safety practices and quality control. The visit will also help us connect our classroom learning with practical engineering and inspire us to think about careers in automation.

We will arrive in a group, carry our ID cards, follow all safety rules, and obey your staff's instructions throughout. We would be glad if a short briefing and a guided tour of about two hours could be arranged.

I would be grateful if you could kindly grant us this opportunity. I look forward to your reply.

Yours faithfully,
Inba Ilakiyan

### Question 3: Letter to the Head of the Mechanical Engineering Department (machine shop access for an electric go-kart)

Inba Ilakiyan
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Head of the Department
Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

**Subject: Request for permission and machine shop access after college hours to build an electric go-kart**

Respected Sir/Madam,

Our student team has been selected to take part in an inter-collegiate design challenge to be held on [date], where we must build and test an energy-efficient electric go-kart. We therefore request permission to use the machine shop after regular college hours.

We need the workshop to cut and weld the chassis, fit the motor, battery and braking system, and test the kart on the campus ground. We estimate that we will need about four weeks, working from 4:30 p.m. to 7:30 p.m. on weekdays and during part of the weekends.

We will work only under the supervision of the workshop instructor, wear safety glasses, gloves and shoes at all times, and follow all machine safety rules. We will take care of the tools and keep the shop clean after use.

We would be thankful if you could grant us this permission and allow us to bring honour to the college.

Thank you for your kind support.

Yours obediently,
Inba Ilakiyan
(On behalf of the team)

### Question 4: Letter to the Principal / Dean of Student Affairs (sponsorship and duty leave)

Inba Ilakiyan
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Principal / Dean of Student Affairs
St. Joseph's College of Engineering
OMR, Chennai – 600 119

**Subject: Request for financial sponsorship and duty leave to present a drone surveillance prototype at the National Defense Innovation Expo**

Respected Sir/Madam,

I am happy to inform you that a five-member multidisciplinary team of our college has been selected to present a drone surveillance prototype at the National Defense Innovation Expo, to be held on [dates] at [venue]. We respectfully request financial sponsorship and duty leave for this event.

Our prototype uses a camera-equipped drone with image processing to monitor sensitive areas in real time. Presenting it at a national level will give us valuable feedback from experts and will bring recognition to our college.

We request sponsorship for travel, accommodation, registration and prototype transport, which we estimate at ₹[amount], and duty leave for [number] days so that we do not lose attendance. We will make up for the missed classes and submit a report after the event.

I would be grateful for your kind support and approval.

Thank you.

Yours obediently,
Inba Ilakiyan
(On behalf of the team)

## Worksheet 2 – Topic 5: Compare and Contrast Descriptive Essays

### Question 1: Electric Vehicles versus Internal Combustion Engine Vehicles

**Electric Vehicles versus Conventional Internal Combustion Engine Vehicles**

Electric vehicles (EVs) and internal combustion engine (ICE) vehicles are two different answers to the same need, which is transport. Each has clear positives and negatives.

Consider battery life first. An EV battery usually lasts eight to ten years, but replacing it is very expensive. In contrast, an ICE vehicle has no battery problem, though its engine needs regular servicing. As for the manufacturing footprint, EVs create more emissions during production because of battery mining and processing, whereas ICE vehicles have a lighter manufacturing footprint. However, over its lifetime an EV

produces far lower emissions, especially when the electricity comes from renewable sources.

Operating costs also differ. EVs have fewer moving parts, so they need less maintenance, and electricity per kilometre is cheaper than petrol or diesel. On the other hand, ICE vehicles cost less to buy, and their repair shops are available everywhere. Fuel prices, however, keep rising and make them costly to run.

Charging network feasibility is the biggest challenge for EVs. Charging stations are still limited, especially on highways and in small towns, and charging takes much longer than refuelling. ICE vehicles, conversely, can be refuelled in minutes at petrol stations spread across the country.

In conclusion, EVs are cleaner and cheaper to run, whereas ICE vehicles are cheaper to buy and more convenient at present. As battery prices fall and charging networks grow, EVs are likely to become the better choice for the future.

## Question 2: Remote/Virtual Internships versus On-Site Internships

**Remote Internships versus On-Site Internships for First-Year Engineering Students**

Internships help early-stage engineering students connect classroom theory with real work. Today, students can choose between remote (virtual) and on-site (physical) internships, and each has advantages and disadvantages.

Hands-on skill acquisition is the biggest difference. On-site internships give direct access to machines, laboratories and real equipment, so students learn practical skills such as wiring, fabrication and testing. Remote internships, in contrast, mostly involve software, simulations and online tasks, which are useful for coding and design but cannot replace physical practice.

In terms of schedule flexibility, remote internships win clearly. Students can work from home, save travel time and cost, and manage studies alongside the internship. On-site internships, on the other hand, demand fixed hours and daily travel, which can be tiring during the semester.

Mentorship is stronger on-site, because supervisors can guide students directly and correct mistakes immediately. Online mentors, however, are often busy and communicate only through scheduled calls or messages, so doubts may stay unresolved for days.

Professional networking also favours on-site internships, as students meet colleagues, build relationships and learn workplace culture naturally. Remote interns may never meet their team face to face, although online platforms still allow them to connect with professionals from different cities.

In conclusion, remote internships are better for flexibility and software skills, whereas on-site internships are better for practical learning, mentorship and networking. A first-year student should ideally begin with a remote internship and move to an on-site one later.

## Question 3: Cloud Computing Infrastructure versus On-Premise Data Centers

**Cloud Computing versus On-Premise Physical Data Centers for Engineering Enterprises**

Engineering enterprises must decide whether to run their systems on the cloud or in their own physical data centers. Both choices have strengths and weaknesses.

Upfront capital expenditure is the clearest difference. Cloud services need almost no initial investment, because companies simply pay for what they use. On-premise data centers, in contrast, require heavy spending on servers, cooling systems, power backup and buildings before any work begins.

As for recurring overheads, cloud bills can grow quickly if usage is not monitored, whereas an on-premise setup has predictable costs once it is built, although it still needs staff, electricity and maintenance. Therefore, cloud suits changing workloads, while on-premise suits stable and long-term demand.

Workload scalability strongly favours the cloud. Companies can add computing power in minutes during simulations or product launches and reduce it afterwards. On the other hand, on-premise systems have fixed capacity, so expanding them takes weeks or months and leaves hardware idle during quiet periods.

Data security and compliance are more complex. On-premise data centers give the company full physical control, which is important for defence, healthcare or confidential design data. Cloud providers offer strong security tools and certifications, but the company depends on a third party and must follow data-location laws

carefully.

In conclusion, the cloud is better for flexibility, speed and low starting cost, while on-premise infrastructure is better for control and predictable long-term expenses. Many enterprises now choose a hybrid model that combines the advantages of both.

### Question 4: 3D Printing versus Traditional CNC Milling

**3D Printing (Additive Manufacturing) versus CNC Milling (Subtractive Manufacturing) in Prototype Fabrication**

In prototype fabrication, 3D printing and CNC milling follow opposite principles. A 3D printer builds a part layer by layer, whereas a CNC machine cuts material away from a solid block. Each method has positives and negatives.

Material wastage is the first point of contrast. 3D printing uses only the material needed for the part, so wastage is low. CNC milling, on the other hand, removes a large amount of material as chips, which increases the cost of expensive metals.

When it comes to production speed, the picture changes with batch size. For single prototypes and small batches, 3D printing is faster because it needs no tooling or setup. For larger batch runs, however, CNC milling is quicker, as it can produce identical parts continuously at high speed.

Surface finish and accuracy favour CNC milling. It produces smooth surfaces and very tight tolerances, so parts often need no extra work. In contrast, 3D printed parts often show layer lines and need sanding or polishing, and their accuracy is lower.

Geometric complexity is the strength of 3D printing. It can easily create hollow parts, internal channels and lattice structures that are impossible to cut with a rotating tool. CNC milling is limited by tool access and the direction of cutting.

In conclusion, 3D printing is ideal for complex, low-volume prototypes with little waste, whereas CNC milling is better for accurate, smooth and larger-batch parts. Engineers often use both methods together to get the best result.

## Worksheet 2 – Topic 6: Personal Email Writing

### Question 1: Email to a high school mentor/teacher about the first semester

**To:** [teacher's email address]

**Subject:** My first semester at engineering college

Dear Sir/Madam,

I hope you are doing well. I wanted to write to you because your guidance in school gave me the confidence to start my engineering journey, and I would like to share how my first semester has been.

The biggest change was the move to independent campus life. Nobody reminds us about deadlines now, so I had to learn to manage my own time, plan my study hours and take responsibility for my work. It was difficult at first, but I am getting used to it.

In our laboratory sessions, we worked on our first small projects, such as basic programs in Python and C and simple circuits. Seeing my own code run correctly gave me a great sense of satisfaction, and I realised how different practical learning is from reading theory.

For the end-semester examinations, I have made a timetable that gives equal time to every subject. I revise the notes every weekend, solve previous question papers and practise the lab experiments so that I do not forget the procedures.

Thank you for everything you taught me. I will be very happy to hear from you and to receive any advice for the coming semesters.

With respect and regards,
Inba Ilakiyan

### Question 2: Email to a high school science teacher about the first term

**To:** [science teacher's email address]

**Subject:** Reflections on my first term in engineering

Dear Sir/Madam,

I hope you are keeping well. It has been a few months since I joined engineering college, and I wanted to share my first-term experience with you, since you were the one who made science exciting for me.

The first thing I noticed was how different the hands-on lab coursework is. In school, we mostly saw experiments in a few periods, but here we handle components, tools and software ourselves. I made mistakes in the beginning, but each mistake taught me something and I am becoming more careful and confident.

I also took part in my first technical club activity and a small hackathon. Working with a team to build a solution within a limited time was challenging and thrilling, and I learned a lot about teamwork and debugging from my seniors.

For the upcoming university theory and practical exams, my strategy is simple. I will finish revising each unit at least two weeks before the exams, practise numerical problems daily, and repeat the lab experiments so that I can answer viva questions without fear.

I would be grateful for any advice you may have. Please take care.

Yours sincerely,
Inba Ilakiyan

### Question 3: Email to an elder sibling/alumnus seeking advice

**To:** [sibling's or alumnus's email address]

**Subject:** Need your advice on balancing studies and technical clubs

Dear [Name],

How are you? I hope college and work are going well. I am writing because I need your advice on something I am struggling with.

My day starts at 6:30 a.m., and I attend classes until 4:30 p.m. After that, I go to my technical club meetings for about two hours, and in the evening I try to study and finish my assignments. By the time I am free, I am usually very tired.

My biggest challenge is managing the lab record submissions. Because the records must be completed every week, they pile up, and I often finish them at the last minute, which affects the quality.

Since you have handled the same situation, could you please tell me how you balanced your academics with the clubs? I would also like your tips for semester exam revision, such as how to plan the timetable, which subjects to start with, and how to use previous question papers.

I am looking forward to your reply. Thank you in advance, and please take care.

Your loving brother/sister,
Inba

## Worksheet 2 – Topic 7: Travel Planning and Budgetary Itineraries

### Question 1: 4-day tour to an aerospace and satellite launch station

**A 4-Day Student Technical Tour to an Aerospace and Satellite Launch Station**

Our department plans a four-day tour to a satellite launch station for forty students and two faculty members. We will travel by a chartered train coach and local bus, so that the cost is reasonable.

**Budget (per student):** train fare ₹1,200, local bus ₹400, accommodation for three nights ₹1,500, food ₹1,200, entry and miscellaneous charges ₹400. The total is ₹4,700, which gives about ₹1,88,000 for the full group of forty.

**Day 1:** Overnight travel and arrival, check-in at the guest house and an orientation briefing. **Day 2:** Visit the launch complex viewing gallery, the rocket assembly area and the mission control centre, where scientists explain how a launch is planned and tracked. **Day 3:** Visit the satellite integration and testing laboratory and the space museum, followed by an interaction with engineers on careers in aerospace. **Day 4:** Morning group discussion, preparation of our tour report, and the return journey.

**Safety protocols:** Carry your ID cards and keep them visible at all times. Follow the instructions of the security staff, avoid photography in restricted areas, and stay with the group. Wear closed shoes and drink plenty of water. Keep the emergency contact numbers of the faculty coordinators and follow the launch-site rules strictly.

This tour will give us a first-hand view of how rockets and satellites are built and operated.

## Question 2: 4-day tour to a major commercial port and maritime dry-dock facility

**A 4-Day Industrial Exposure Tour to a Major Commercial Port and Dry-Dock Facility**

Our class of forty first-year students and two faculty members will visit a major commercial port and a dry-dock facility over four days. We will travel in a hired bus, and stay in a budget hotel near the port.

**Estimated cost per student:** travel ₹1,500, stay for three nights ₹1,800, food ₹1,200, entry passes and miscellaneous ₹500. The total is ₹5,000 per student.

**Day 1:** Travel to the port city, check in and a short orientation. **Day 2:** Technical briefing at the port authority, followed by a visit to the container terminal, where cranes, conveyors and tracking systems handle cargo. **Day 3:** Visit the dry-dock, where ships are lifted and repaired, and see how hull cleaning, welding and engine maintenance are done. An engineer will also brief us on ship stability. **Day 4:** Visit the control tower and the vessel traffic system, a feedback session and the return journey.

**Dockyard safety protocols:** Wear helmets, safety shoes and high-visibility jackets at all times. Do not go near moving cranes, cargo or the edge of the dock. Stay in the group, obey the guides and never touch machinery. Keep emergency contacts ready, and carry first-aid supplies.

This tour will help us understand how ports and shipyards work in real life.

## Question 3: 4-day tour to an electric locomotive manufacturing works and railway signalling center

**A 4-Day Technical Study Tour to an Electric Locomotive Works and a Railway Signalling Center**

We plan a four-day study tour for thirty-five students and two faculty members to an electric locomotive manufacturing works and an automated railway signalling center. We will book sleeper-class seats on trains for both journeys, and use a local bus for the visits.

**Category-wise expenditure per student:** rail tickets ₹1,400, local transport ₹300, accommodation for three nights ₹1,500, food ₹1,200, and entry and miscellaneous ₹300. The total is ₹4,700.

**Day 1:** Overnight train journey and arrival. **Day 2:** Visit the locomotive works, where we will see the shop-floor stages such as shell fabrication, wiring, traction motor assembly and final testing of the engine. **Day 3:** Visit the automated signalling center, where engineers explain how sensors, track circuits and computers control train movement and keep the trains safe. **Day 4:** Interaction with engineers, preparation of a tour report and the return journey.

**Site safety rules:** Wear helmets, safety shoes and closed clothing in the shop floor areas. Never walk on the railway tracks, and keep a safe distance from cranes and live electrical equipment. Follow the guides' directions and stay with the group. Do not touch any machine or signal equipment, and report any problem immediately to the faculty coordinator.

This tour will connect our classroom learning with real railway engineering.

# Project Proposal: Ideation

LunArk: Genesis, Technical Specification

Team LunArk | NASA Space Apps Challenge 2026 | Draft v1, October 17, 2026

Status: draft. Items marked [default] are working assumptions the team can change. Items marked [open] still need a decision.

## 1. Overview

LunArk: Genesis is a desktop game where the player runs a lunar outpost. You place buildings, keep power flowing through the long lunar night, protect the base from radiation, earn money and stop the crew from getting bored. The aim is to let students aged 13-16 feel the engineering trade-offs that decide whether a real Moon mission succeeds or fails.

The game is built in Python with pygame-ce. All art is hand-drawn pixel art made in Aseprite. The code is written with help from an AI coding agent (Google Antigravity) and reviewed by the team, and the project submission discloses this.

## 2. Goals and non-goals

### 2.1 Goals

- Make real mission trade-offs tangible: power, life support, radiation shielding, money and crew morale.
- A 13-16 year old should understand it within about ten minutes, thanks to a guided tutorial.
- It runs on Windows, macOS and Linux with two commands and no installer.
- It is fully documented, so a judge or a new teammate can run, read and extend it.
- Game logic, user interface and game data stay strictly separate.

### 2.2 Non-goals

- No multiplayer, online features or accounts.
- No packaged executables (no PyInstaller). Players run the game from source.
- No AI-generated art, fonts or audio. The team makes all art.
- No zooming or rotating of sprites, which keeps the pixel art crisp.
- No real orbital mechanics. The science content is simplified, but it must be accurate.

## 3. Target audience and design feel

- Audience: students aged 13-16, with little or no engineering background.
- Feel [default]: calm and hopeful at the start, with pressure rising later, but never unfair. When players fail, they should think “I see what I did wrong”, not “that was random”. Failure screens always show the exact reason and a hint.
- Inspiration: the mobile game Spaceflight Simulator, used only for ideas. Those are a clean, flat, icon-first interface; building from parts with a ghost preview; a side-view 2D world; an always-visible time warp that slows itself down when something important happens; and predictive information such as countdowns and warnings. No art, code, name or exact UI is copied.

## 4. Technology stack

| Item | Decision |
|---|---|
| Language | Python 3 (minimum tested version to be recorded in the README) |
| Library | pygame-ce, version pinned in requirements.txt |
| Testing | pytest (development only, in requirements-dev.txt) |
| Art tool | Aseprite, exported as PNG |
| Packaging | None. Run from source |
| Platforms | Windows, macOS, Linux, desktop only |

### 4.1 How players run the game

```
pip install -r requirements.txt
python3 main.py
```

On Windows the second command may be python main.py or py main.py. The README documents all three, along with optional virtual-environment steps.

## 5. Repository structure

```
LunArkGenesis/
  main.py                entry point
  requirements.txt       pygame-ce only, pinned
  requirements-dev.txt   pytest
  README.md              full documentation
  LICENSE                GPL-3.0
  fonts/                 pixel font (team-supplied)
  img/                   all art (team-made)
    logo/  ui/  icons/  terrain/  sky/  buildings/
    characters/  cutscene/  events/  fx/  vehicles/
  saves/                 runtime only, gitignored
  tests/                 pytest tests for game/logic
  game/
    app.py               window, main loop, screen manager
    constants.py         fixed numbers, colors, paths, index tables
    assets.py            image/font loading, placeholders, 9-slice
    logic/               game rules, no pygame imports
    ui/                  drawing and input
    screens/             splash, title, settings, cutscene, play, pause, end
    data/                fixed game content as JSON
```

Rule of thumb: logic decides what happens, ui shows it, data holds the numbers and text, and saves holds the player’s files.

## 6. Architecture

### 6.1 Layers

| Layer | Responsibility | Rules |
|---|---|---|
| game/logic | Bars, clock, world grid, buildings, events, win/lose, settings, save/load | Never imports pygame. Fully unit-tested |
| game/ui | Screens, widgets, drawing, input, display scaling | Contains no game rules |
| game/data | buildings.json, events.json, balance.json, story.json | Numbers and text only |
| game/assets.py | Loads images and fonts | Draws a labelled placeholder when a file is missing |
| saves/ | settings.json, savegame.json | Created at runtime, validated on load |

### 6.2 Main loop

The app owns one window and a screen manager. Each screen implements handle events, update and draw. The simulation moves forward in fixed logic ticks (1 tick = 1 second at 1x speed), independent of the frame rate, so the FPS setting never changes game speed.

### 6.3 Error handling

A missing or corrupt save, settings file, JSON file or image must never crash the game. It falls back to defaults or placeholders and logs a clear message. At startup it prints a single summary of missing assets.

## 7. Display, fullscreen and settings

### 7.1 Rendering

- The game draws to an internal canvas of 480 x 270 [default] (16:9) with 16 x 16 pixel tiles.
- The canvas is scaled to the window or screen with nearest-neighbor scaling. The aspect ratio is kept, and black bars fill any gaps.
- Mouse positions are converted back to canvas coordinates.
- An optional integer scaling setting uses whole-number scale factors only, which gives sharper pixels but larger bars. Default: on.

### 7.2 Display modes

- The first launch starts in fullscreen at the desktop resolution, and the choice is remembered.
- F11 toggles between fullscreen and windowed mode from any screen.
- Esc never quits instantly. It goes back in menus and opens the pause menu in the game. Quit is a button on the title and pause menus.
- On Windows the process is made DPI-aware so sizes are correct on scaled displays.

### 7.3 Settings menu

Available from the title screen and the pause menu, by mouse or keyboard.

| Setting | Options |
|---|---|
| Display mode | Fullscreen, Windowed |

| Setting | Options |
|---|---|
| Resolution (windowed size) | 1280x720, 1366x768, 1600x900, 1920x1080, 2560x1440, Native desktop. Only sizes that fit the monitor are listed. Greyed out in fullscreen |
| FPS cap | 30, 60, 120, 144, Unlimited. Default 60 |
| VSync | On, Off. Default on. Hidden if unsupported |
| Show FPS counter | On, Off. Default off |
| Integer scaling | On, Off. Default on |
| Buttons | Apply, Back, Reset to defaults |

Changing the display mode or resolution brings up a “Keep these settings? (reverting in 10s)” prompt that reverts on its own, so nobody gets stuck on a bad screen. Settings are saved to saves/settings.json, validated on load, and replaced by defaults if invalid.

## 8. Screen flow

1. Splash: a black screen. The logo (33 x 33 px, integer-scaled with nearest-neighbor) fades in and out over about 3 seconds. Any key or click skips it.
2. Title screen: wordmark and background. Menu: New Game, Continue (disabled without a save), Settings, Quit.
3. Intro cutscene: a full-screen picture and a dialogue box with typewriter text. The player wakes up to a man looking at them. He welcomes them to the Moon and asks their name (typed, maximum 12 characters, default “Commander”). He explains that NASA sent them to establish a base and introduces himself as Lieutenant Hale [default name], then walks through the first steps. The cutscene can be skipped.
4. Guided tutorial: inside the real game. The lieutenant waits for the player to finish each step: place a Fuel Cell Generator, place the Habitat Pod, watch the Energy bar, pan the camera, build a Lab, understand Boredom and build a Gym, understand the Event bar, read the sunset countdown, then dig a building underground and see its radiation exposure improve.
5. Main game.
6. Pause menu: Resume, Settings, Save, Quit to Title, Quit Game.
7. End screens: win or lose, always with the exact reason and a hint for next time. Buttons: Retry, Title.

## 9. Game design

### 9.1 The world

- A side-view cross-section of the Moon, 48 columns wide [default], scrolling horizontally on the 480 px canvas.
- From top to bottom: black sky (the lunar sky is black even in daytime), the ground line, then 4 rows [default] of underground regolith.
- The sky shows stars, the Sun crossing during the day, and Earth hanging almost still. Two parallax horizon layers add depth.
- The camera pans horizontally only (A and D, arrow keys, or dragging with the middle mouse button). There is no zoom.

### 9.2 Placing buildings

- The player picks a building from a collapsible build tray at the bottom of the screen.
- A ghost preview follows the mouse, snapped to the grid. It turns green when the spot is valid and red when it isn't, with a one-line reason such as "Needs the surface", "Not enough cash", "Locked: needs Science 50" or "Space is taken".
- Left click places, and right click or Esc cancels. Demolishing refunds part of the cost.
- Each building has a placement type: surface, underground or either.
- Building underground digs the footprint automatically. The cost is the building cost plus a dig cost per tile. No access shafts are modeled, which is a deliberate abstraction stated in the README.
- Radiation exposure runs from 1.0 on the surface down to about 0.05 when deep, depending on how many regolith rows sit above the building. A small radiation icon shows it.
- Selecting a building opens an info card with its role, energy in and out, exposure, status, a science fact, and Repair or Demolish buttons.

### 9.3 Time and day-night cycle

- 1 tick equals 1 second at 1x speed. Time warp buttons: Pause, 1x, 2x, 4x, 8x.
- Warp drops back to 1x automatically, with an on-screen reason, when a warning or event occurs or when Energy falls below 15% [default].
- The real lunar night lasts about 14 Earth days, so it is compressed: 180 ticks of sunlight, then 180 ticks of night [default], set in balance.json.
- The HUD shows a sun or moon icon and a countdown to the next sunset or sunrise.

### 9.4 The five bars

| Bar | Meaning |
|---|---|
| Energy | Stored energy between 0 and a capacity. Each tick: production minus consumption. At 0, buildings shut down in priority order, with the Habitat Pod last |
| Science | Accumulated research points. Used for the win goal and for unlocking equipment tiers |
| Boredom | 0 to 100. Rises over time with crew size. Activity buildings lower it |
| Revenue | Cash balance. Spent on building, digging and repairs, earned by Cashflow buildings. Shown as a bar toward the next unlock plus a number |
| Event | 0 to 100. Drifts up when Boredom is high and down when Boredom is low. A higher value means a higher chance per tick of a random failure event |

Bars flash when they run low or turn dangerous. All rates and thresholds live in balance.json, so they can be tuned without code changes.

### 9.5 Buildings

There are four categories with three tiers each. Tiers 2 and 3 unlock only when both Science and Revenue pass thresholds set in balance.json.

| Category | Tier 1 | Tier 2 | Tier 3 | Rule |
|---|---|---|---|---|
| Generator | Fuel Cell Generator (2x2) | Solar Array (3x2) | Nuclear Plant (3x3) | Produces Energy |
| Cashflow | Datacentre (2x2) | Precious Metal Mining (3x2) | Helium-3 Extraction (3x3) | Uses Energy, produces cash |

| Category | Tier 1 | Tier 2 | Tier 3 | Rule |
|---|---|---|---|---|
| Activity | Gym (2x2) | Cafe (2x2) | Restaurant (3x2) | Uses Energy, lowers Boredom |
| Science | Lab (2x2) | Land Research (3x2) | Planet Stats (3x3) | Uses Energy, produces Science |

Special building: the Habitat Pod (3x2) provides life support and housing. It must stay powered, and it is placed during the tutorial.

Default placement: surface for Fuel Cell Generator, Solar Array, Precious Metal Mining, Helium-3 Extraction, Land Research and Planet Stats. Either for Habitat Pod, Datacentre, Gym, Cafe, Restaurant, Lab and Nuclear Plant.

Science accuracy rules:

- The first generator is a fuel cell, not a diesel generator. Burning fuel needs oxygen, and the Moon has no air. The fuel is shipped from Earth, so it carries a running cash cost per tick.
- The Solar Array produces nothing at night, and dust storms reduce its output. That is the central trade-off against the expensive, constant Nuclear Plant.
- The Nuclear Plant has high constant output and a very high cost, and it is locked behind Science.
- The lunar resource is Helium-3, not Helium II.
- Every building and event carries a short science fact written for ages 13-16, flagged verified: false until the team checks it against NASA sources.

### 9.6 Events

Each event has an id, name, description, effect, minimum Event bar level, warning time, preparation tip and science fact.

| Event | Effect |
|---|---|
| Solar Flare | Warning about 20 ticks ahead. Each building rolls a chance of going offline equal to a base chance times its radiation exposure. A surface Habitat Pod makes the crew shelter, pausing Science and spiking Boredom. Buried buildings are mostly safe |
| Dust Storm | Solar Array output halved for some ticks |
| Equipment Failure | One random building goes offline until repaired |
| Micrometeorite | Small cash loss |
| Crew Argument | Boredom spike |

Warnings show a banner with a countdown. Events show a popup that says what happened, why, and a tip. Repairs cost cash and are done from the info card.

### 9.7 Win and lose conditions [default]

- Win: build Planet Stats and keep it running for a short period.
- Lose, life support: the Habitat Pod stays unpowered for more than a set number of ticks.
- Lose, stranded: zero Energy, zero cash and no way to earn either.

### 9.8 HUD layout

Fitted to 480 x 270. The five bars with icons run along the top left. Time warp and the sun or moon countdown sit at the top right, next to a settings gear. A collapsible build tray at the bottom center has five category tabs (Generator, Cashflow, Activity, Science, Habitat). The info card appears on the right when something is selected, and a small message log sits at the bottom left. Icons come first and text second, with tooltips on hover.

## 10. Data files

All game content lives in game/data as JSON, so it can be tuned without touching code.

| File | Contents |
|---|---|
| buildings.json | id, name, category, tier, cost, placement, footprint, energy use or production, cash per tick, science per tick, boredom reduction, sprite, description, science fact, verified flag |
| events.json | id, name, description, effect, minimum Event bar, warning ticks, preparation tip, science fact, verified flag |
| balance.json | day and night lengths, bar rates, unlock thresholds, dig cost per tile, refund rate, radiation exposure table, warp auto-slow threshold, win and lose thresholds |
| story.json | all cutscene and tutorial lines, mentor name, placeholders for the player’s name |

Building ids, which are also the sprite file names: habitat_pod, fuel_cell_generator, solar_array, nuclear_plant, datacentre, precious_metal_mining, helium3_extraction, gym, cafe, restaurant, lab, land_research, planet_stats.

## 11. Art pipeline

All art is hand-drawn in Aseprite and exported as PNG. The loader accepts any file placed at the documented path, with no code change. Animated sprites are horizontal strips of equal-size frames, and the frame count is detected automatically from the image width. For buildings, frame 0 is offline and later frames are the running animation. Missing art shows a labelled placeholder of the correct size.

| Area | Path and size |
|---|---|
| Logo | img/logo/logo.png, 33x33 |
| UI pieces | img/ui/: 9-slice panel (24x24), button (4 states, 96x24), bar frame, dialogue box, toggle, next arrow, cursor, title, title background, end screens |
| Icons | img/icons/icons.png, one 128x48 sheet of 24 icons at 16x16 in a fixed order |
| Terrain | img/terrain/tileset.png, 128x48, 24 tiles at 16x16 |
| Sky | img/sky/: stars, earth, sun, far and near horizon |
| Buildings | img/buildings/<id>.png, frame size equals footprint times 16 px |
| Characters | img/characters/mentor.png (4 expressions at 64x64), astronaut (optional) |
| Cutscene and events | img/cutscene/ at 480x270, img/events/ at 96x64 |
| Effects and vehicles | img/fx/, img/vehicles/ (optional) |
| Font | fonts/*.ttf, a pixel font whose license has been checked and credited |

The separate Asset Guide describes every sprite, its size, its priority (must, should, nice) and a suggested drawing order. About 15 files make a playable demo.

## 12. Testing and quality

- Every rule in game/logic has a pytest test, and tests must pass before a phase is finished.
- A terminal runner simulates a game as text, so the balance can be tuned before any graphics exist.
- Each phase has a manual test checklist: fullscreen, F11, resolution change, the auto-revert confirmation, and running with no art files at all.
- An F3 debug overlay lists which assets are still placeholders.

## 13. Documentation requirements

The README is written and updated after every phase. It contains a description and screenshot, a 30-second quick start, controls and settings, how to play, the science behind the game, a plain-language tour of every folder and file, the architecture, how-to guides (add a building, event, balance change, art, font or screen), data schemas, how to run tests, the art guide, the AI disclosure, known issues, roadmap, changelog, credits and license.

All code has type hints and docstrings, and comments explain why, not what.

## 14. Use of AI and disclosure

- Code is written with the AI coding agent Google Antigravity, one phase at a time, and the team reviews it before the next phase starts.
- Every Python file carries a header comment saying it is AI-assisted.
- All art is made by hand in Aseprite. The agent is forbidden from creating image, font or audio files.
- The README and the Space Apps submission form say where AI was used, as the event requires.
- The team must be able to explain every file to the judges.

## 15. Development plan

| Phase | Content | Target |
|---|---|---|
| 1 | Scaffold, display system, settings menu, splash, title screen, README v1 | by Oct 22 |
| 2 | Headless game logic, data files, tests, terminal runner, balance tuning | by Oct 28 |
| 3 | Side-view play screen: world, camera, build tray, ghost preview, digging, HUD, time warp, info card, save and load | by Nov 4 |
| 4 | Intro cutscene, name input, guided tutorial | by Nov 8 |
| 5 | Warnings, event popups, forecast, end screens, polish, debug overlay, QA, final README | by Nov 11 |
| Freeze | No new features. Demo video, final README, submission | Nov 12-13 |
| Event | Space Apps weekend | Nov 14-15 |
| 6 (stretch) | Resupply lander, minimap, battery building, sound hooks, extra events | only if time allows |

Art is drawn in parallel, starting with the minimum set. If time runs short, the full-scale game is cut back from Phase 5 and Phase 6 first.

## 16. Risks and mitigations

| Risk | Mitigation |
|---|---|
| Event rules may not allow work done before the event [open] | Ask the local organizer. If pre-building is not allowed, use the time for planning, art and learning, and build during the 48 hours |
| Scope is too large for the time | Phased plan, strict freeze date, cut stretch items first |
| The AI agent produces code the team does not understand | One phase at a time, read every diff, ask the agent to explain files |
| Balance feels wrong | Data-driven numbers and a text-only simulation before the UI |
| Art is not ready in time | Placeholders keep the game playable; draw the minimum set first |
| Judges cannot run the game | Clear quick start, demo video, tested on a clean machine |
| Fullscreen gets stuck on someone’s machine | F11 and Esc always work, and display changes auto-revert |
| Science facts are wrong | Mark all facts unverified until checked against NASA sources |
| AI usage limits are reached mid-build | Keep a backup agent or local model ready, commit after every working step |

## 17. Open questions

1. Is work done before the event allowed? [open, confirm with the organizer]
2. Are time and tile sizes final (480 x 270 canvas, 16 x 16 tiles)?
3. Is the game real-time with time warp (the current default) or turn-based?
4. What is the mentor’s final name?
5. What are the final win and lose rules and balance numbers?
6. Which pixel font will be used, and under what license?

## 18. Glossary

- Regolith: the loose dust and rock covering the Moon’s surface.
- Tick: one step of the game simulation, equal to one second at 1x speed.
- 9-slice: a way to stretch a UI image without distorting its corners.
- Exposure: how much radiation reaches a building, from 1.0 (surface) down to about 0.05 (deeply buried).
- Ghost preview: the see-through building that follows the mouse before it is placed.
- Time warp: speeding up the simulation clock.

# LinkedIn and GitHub Profiles

## Part A: LinkedIn profile

## Inba Ilakiyan

Aspiring Principal Cloud Architect and eventually Director of Engineering

CSE Student (Batch 2026–2030) | Python, C & Linux | github.com/DubberRuckky

### About

I'm a knowledge-hungry Computer Science Engineering student, and I'm willing to adapt and improvise to get past any problem. Most of my work is in Python, C and Linux. I like to think a challenge all the way through before I build anything.

Ideally I'd become a game developer. Starting in infrastructure development or cloud architecture would give me the skills I need to get there, and those roles ask for more situational awareness than an algorithm is likely to replicate. My long-term goal is to become a Principal Cloud Architect and eventually a Director of Engineering.

Fun fact: I built my first PC at 15.

Focus areas: Linux • Python • C • Game development • Cloud fundamentals • Systems thinking

### Featured Projects

**Royal-Roll**

A casino simulator and dice roller. It explores how a casino operates and what lies behind the saying "the house always wins".

Tech: Python | github.com/DubberRuckky/Royal-Roll

**latte**

A hobby game project. I'm learning Pygame through it, to become a better game developer and a better coder.

Tech: Python, Pygame, Game Development | github.com/DubberRuckky/latte

### Education

Computer Science & Engineering (CSE), 1st Year. Batch 2026 – 2030.

### Honors & Awards

- 1st Place, Mazebot Challenge – Sathyabama Technosummit '26
- Overall Shield Winner (Regional Level) – AWIM 2018

### Skills

| | |
|---|---|
| Languages | Python, C |
| Operating Systems | Linux, Windows |
| Tools | Git, GitHub, Pygame (learning) |
| Strengths | Problem solving, adaptability, improvisation, self-directed learning |

### Licenses & Certifications (Planned)

- AWS Certified Cloud Practitioner – target: before Year 2
- AWS Certified Solutions Architect – Associate – target: Year 2
- CKA or HashiCorp Terraform Associate – target: Year 3

### Career Roadmap

| Year | Focus | Deliverable |
|---|---|---|
| Year 1 (2026–27) | Linux, networking, cloud basics | Portfolio site on AWS S3 / CloudFront |
| Year 2 (2027–28) | Docker, Python automation, IaC | 2-tier VPC deployed with Terraform |
| Year 3 (2028–29) | Kubernetes, CI/CD, DevSecOps | Microservices on EKS via GitHub Actions |
| Year 4 (2029–30) | SRE, FinOps, leadership | Multi-cloud disaster recovery capstone |

### Connect

GitHub: github.com/DubberRuckky | LinkedIn: linkedin.com/in/inba-ilakiyan-a-e-51909742a

## Part B: GitHub profile README

# Hi there, I’m Inba Ilakiyan

CSE Student | Linux Enthusiast | Aspiring Principal Cloud Architect & Director of Engineering

LinkedIn: https://www.linkedin.com/in/inba-ilakiyan-a-e-51909742a/
GitHub: https://github.com/DubberRuckky
LeetCode: https://leetcode.com/u/Inba_Ilakiyan/

### About Me

- I’m a Computer Science Engineering student, building a foundation in software, systems and the cloud.
- I’m working toward becoming a Principal Cloud Architect and, longer term, a Director of Engineering.
- Proud Linux enthusiast: I like living in the terminal and understanding how systems work under the hood.
- Fun fact: I built my first PC at 15.

### Tech Stack & Tooling

| Domain | Technologies & Tools |
|---|---|
| Languages | Python, C |
| Operating Systems | Linux, Windows |
| CI/CD & Version Control | Git, GitHub |

### Featured Repositories

**Royal-Roll** (https://github.com/DubberRuckky/Royal-Roll)

A casino simulator and dice roller. It shows how a casino operates and what the saying “the house always wins” means.

Python

**latte** (https://github.com/DubberRuckky/latte)

A hobby game project. I’m learning Pygame to become a better game developer and a better coder. Python • Game Dev

### Career Roadmap

- Done: build strong CS fundamentals and Linux skills.
- In progress: learn cloud platforms, containers and Infrastructure as Code.
- Goal: earn foundational cloud certifications (e.g. AWS Cloud Practitioner / Solutions Architect Associate).
- Goal: grow into cloud architecture, then technical leadership.

### Certifications & Achievements

- 1st place in Sathyabama Maze Bot Challenge. Overall Shield Winner (Regional Level) in AWIM 2018.

### Connect With Me

I’m always happy to connect with fellow students and engineers interested in cloud and Linux. Reach out on LinkedIn: https://www.linkedin.com/in/inba-ilakiyan-a-e-51909742a/